

Table 1: Kuo Quadrupole F20 (fitting region: cube $\pm 70 \mu\text{m}$, full FEM mesh)

$$\text{Error} = \frac{1}{N} \sum_{i=1}^N \frac{|\mathbf{B}_{model} - \mathbf{B}_{FEM}|_i}{|\mathbf{B}_{FEM}|_i} \cdot 100$$

Kuo Quadrupole (Error=7.11%)		
Pole	$ \ell $ [μm]	Tip→Charge [μm]
P1	502.8	2.8
P2	502.7	2.7
P3	502.8	2.8
P4	503.0	3.0
Average	502.8	2.8
R_a	$1.54 \times 10^9 \text{ A/Wb}$	

Tip→Origin = 500 μm

Tip→Charge = $|\ell| - 500$

$N_c = 4$ (4 turns per pole), 4-fold symmetric quadrupole, full-mesh 3D fit ($N = 17772$ FEM nodes total). F20 fillet = 20 μm .